

Deep Research Report on Cross-References Between 46.875 Hz, Satellites, Kyndryl, IBM, Jehovah's Witnesses, Setecom, Costa Rica, Huawei, SDA and DARPA Programmes, LeoLabs, PLC, Li-Fi, and Digital Twin Technologies

Executive Summary

This research found several **strong, well-documented link clusters**—but also **clear gaps** where no credible source explicitly connects certain items (notably **46.875 Hz** to satellites or the named defence/telecom organisations).

The **strongest explicit connections** identified are:

First, official U.S. defence sources show a direct chain from the Defense Advanced Research Projects Agency ¹ Blackjack programme to the **Mandrake 1 / Mandrake 2** risk-reduction missions and to proliferated LEO networking concepts, explicitly citing partnership or stakeholder involvement with the Space Development Agency ² . ³

Second, primary Space Development Agency ² material and U.S. Space Force ⁴ releases outline the **Proliferated Warfighter Space Architecture (PWSA)** as a multi-tranche LEO constellation using (among other technologies) **optical inter-satellite links** and **radio-frequency links**, with published details for Tranche 0 and Transport Layer concepts. ⁵

Third, there is an unusually explicit linkage between **space programmes and digital twins**: Space Development Agency ² small-business materials list “**space system digital twin development and integration**” as an area of interest, and Sedaro ⁶ + SBIR documentation describe building **digital twins** for both **GPS satellites** and the **PWSA constellation**. ⁷

Fourth, there are multiple primary/official references tying Kyndryl ⁸ and IBM ⁹ operations to Costa Rica ¹⁰ and specifically Heredia ¹¹ (e.g., corporate entity/billing addresses; workforce and investment announcements; and an official government communiqué about IBM's security operations in Heredia). ¹²

Fifth, Costa Rica's **5G cybersecurity regulation** appears in an official legal repository as **Decreto Ejecutivo 44196 (25/08/2023)**; multiple news sources interpret the framework as excluding bidders from countries not tied to the Budapest cybercrime convention, which is reported to affect Chinese vendors such as Huawei ¹³ (as a practical procurement gate). ¹⁴

Sixth, LeoLabs ¹⁵ has a documented footprint in Costa Rica via an official company press release describing its **Costa Rica Space Radar** as operational and aimed at LEO object tracking; trade press also references the company's Costa Rica radar facility. ¹⁶

The **largest gaps** (important for evidentiary reasoning) are:

No primary or credible secondary source found in this pass explicitly links **46.875 Hz** (or "48000/1024 FFT bin") to **satellites, LEO constellations, DARPA Blackjack, SDA PWSA, LeoLabs, Kyndryl/IBM operations, Costa Rica 5G procurement, or Huawei**. Instead, 46.875 Hz is strongly documented as a **generic DSP artefact** of a 48 kHz sampling rate with a 1024-point FFT, and separately as an **AAC audio frame-rate consequence** (1024 samples per frame at 48 kHz). ¹⁷

No credible source found that *directly* connects **Jehovah's Witnesses** or **Setecom** to the SDA/DARPA/LEO constellation cluster. The strongest link here is geographic: both appear in/near Heredia, Costa Rica, but in unrelated contexts (religious organisation contact details; electrical generator and telemetry business activities). ¹⁸

Terminology ambiguity is a recurring risk: "SDA" often means **Space Domain Awareness** in space-safety contexts, but the user's list includes **Space Development Agency**; "PLC" can mean **power line communication** or **programmable logic controller**, and some Costa Rica sources use "PLC" in a vendor/brand or corporate abbreviation context rather than communications-over-mains wiring. ¹⁹

Research Method and Scope

This report prioritised:

Primary and official documents (government/agency websites, official legal repositories, official press releases, standards and vendor qualification notices) and peer-reviewed technical papers.

English and Spanish sources were prioritised; a smaller number of Chinese-language sources were retained where they explicitly referenced key entities (e.g., PWSA/SDA mentions in Chinese commentary). ²⁰

"Explicit mention" was interpreted literally: the source text must contain **at least two** of the target terms or clearly named entities/technologies from the user list (e.g., "Mandrake 2" + "Blackjack" + "SDA"; "digital twin" + "power line communication"; "Kyndryl" + "Heredia" + "Costa Rica"). Sources that mention only one term were generally excluded unless they materially anchored a high-value chain (e.g., the official Costa Rica 5G decree text). ²¹

Credibility levels used in the table:

Very High: official government/agency legal or programme documentation; official corporate qualification notices; official legal repositories.

High: peer-reviewed journals; respected public technical repositories and public SBIR summaries.

Medium: trade press; company PR (informative but self-interested); well-established local media.

Low: job boards, social media posts, forums (useful for linkage discovery; not strong proof of operational facts without corroboration).

Evidence Clusters and Link Strength Assessment

The 46.875 Hz reference is strongly rooted in audio DSP, not space systems

A primary manufacturer technical note explains that FFT “spectral line” spacing equals sampling rate divided by FFT size, giving $48000/1024 = 46.875$ Hz at 48 kHz with a 1024-point FFT. ²²

Independent developer documentation likewise describes the same bin width in the context of FFT processing at 48 kHz and 1024 samples. ²³

Separately, a widely cited technical forum answer explains that AAC audio (as produced by FFmpeg’s encoder) uses **1024 samples per audio frame**, leading to $48000/1024 = 46.875$ audio frames per second at 48 kHz. ²⁴

Across the sources reviewed here, none used “46.875 Hz” to describe a satellite modulation scheme, LEO network timing, or a defence programme signature; it consistently appears as an arithmetic consequence of **48 kHz** sampling and **1024-sample** blocks (FFT or codec framing). ¹⁷

DARPA Blackjack and Mandrake missions explicitly connect to SDA and proliferated LEO networking

An official DARPA news release (May 11, 2020) states that Blackjack—“in partnership with the U.S. Space Force and Space Development Agency”—targets flights to LEO and aims to demonstrate constellation autonomy and space mesh networks. The same release explicitly names Mandrake 1 and Mandrake 2 and describes their intended payloads (including optical inter-satellite links for Mandrake 2) as enabling optically meshed computer networks in LEO. ²⁵

A later official DARPA news release (Jul 7, 2021) reports successful deployment of two Mandrake 2 satellites (Able and Baker) and explicitly lists the stakeholder team including the Space Development Agency and other U.S. defence organisations, framing Mandrake 2 as risk reduction for Blackjack and focused on advanced laser communications terminals. ²⁶

These two sources provide the most direct “programme-name bridge” in the full term list provided by the user: **Blackjack ↔ Mandrake ↔ SDA ↔ LEO mesh / optical comms.** ³

SDA PWSA documentation provides technical substance for the proliferated LEO constellation concept

An SDA Tranche 0 fact sheet describes Tranche 0 as a “warfighter immersion tranche,” gives counts for Transport and Tracking satellites, and states Tranche 0 vehicles are distributed in two orbital planes at ~1000 km altitude and include optical communications terminals and RF capability. ²⁷

SDA's Transport Layer materials describe a proliferated constellation of hundreds of space vehicles in LEO with optical inter-satellite links via optical communication terminals, alongside RF links, and position this layer as a backbone for military communications concepts. ²⁸

A U.S. Space Force news release (Apr 3, 2023) announces the initial launch of SDA Tranche 0 satellites and reiterates the intent: low-latency data transport and missile warning/tracking from LEO, and optically connected Transport and Tracking satellites in two orbital planes at ~1000 km. ²⁹

Digital twins appear as an explicit theme in U.S. space acquisition and satellite test ecosystems

SDA's publicly posted small-business focus areas include "space system digital twin development and integration," explicitly associating "digital twin" work with SDA's ecosystem and solicitations. ³⁰

A Sedaro announcement (Sep 12, 2024) states that the company received a Direct-to-Phase II SBIR from SDA and will implement a digital twin platform supporting both GPS and PWSA, including building digital twins of GPS satellites and digital twins of the PWSA constellation for operational/training applications. ³¹

An SBIR portfolio description further discusses development of a "PWSA Digital Engineering Software Library" leveraging a "PWSA Digital Twin" to assess performance and resilience of PWSA architectures, explicitly naming SDA and PWSA. ³²

A key analytical point: these are some of the few places where "digital twin" and "constellation/LEO/satellites" appear together in the same U.S. programme context, distinct from industrial or energy-sector usage. ³³

Costa Rica and Heredia have multiple independent anchors across Kyndryl, IBM, Huawei, and LeoLabs

Kyndryl's own procurement invoicing instructions list a Costa Rica legal entity and a billing address in Heredia (Zona Franca Los Arrallanes, Edificio F-31), linking Kyndryl + Costa Rica + Heredia in an official corporate operational context. ³⁴

An IBM-hosted "IBM and Kyndryl Client Guide to Operational Changes" PDF also lists "Kyndryl Costa Rica" with the same Heredia location and tax registration format, reinforcing the same linkage from a separate, high-credibility corporate documentation channel. ³⁵

A CINDE press release (Nov 4, 2021) states Kyndryl completed its separation from IBM and launched operations from Costa Rica via a customer innovation centre, directly connecting Kyndryl + IBM spinoff context + Costa Rica. ³⁶

A Costa Rica government communication (COMEX, Apr 4, 2019) states IBM expanded a global security services centre located in Heredia (America Free Zone) and describes its international coverage footprint, linking IBM + security operations + Heredia, Costa Rica. ³⁷

For Huawei, a Costa Rica legal repository entry shows the 5G cybersecurity regulation as Executive Decree 44196 and describes the regulation's purpose as establishing cybersecurity measures for 5G and higher telecom services. ³⁸ A separate Spanish-language business outlet reports Huawei opened service centres including one in Heredia (commercial retail/service context, not network infrastructure). ³⁹

For LeoLabs, a company press release (Apr 22, 2021) declares its Costa Rica Space Radar “fully operational” and explicitly frames it as enabling improved tracking of LEO objects including active satellites and debris, and describes government officials attending a ceremony in Costa Rica. ⁴⁰ Trade press also references “LeoLabs’ Costa Rica Radar Facility opened in 2021.” ⁴¹

The Costa Rica 5G cybersecurity rule is official; Huawei and China linkage is largely described through reportage and interpretation

The Costa Rica legal text identifies the regulation as “Reglamento sobre medidas de ciberseguridad aplicables” for 5G and higher and gives the decree number and date, providing the primary anchor for timing and legal existence. ³⁸

Multiple news sources describe the practical procurement effect as excluding vendors from countries not aligned with the referenced Budapest framework, and name China and Huawei as impacted; later reporting discusses court action suspending implementation (procedural posture varies by source). ⁴²

Because this linkage is partly interpretive in media reporting (how procurement eligibility translates to real vendor exclusion), the most defensible approach is to treat the decree as the primary source and treat “Huawei excluded” as a reported consequence, not a self-evident fact from the decree text alone. ¹⁴

PLC and Li-Fi are well-documented together, but primarily in indoor/power and industrial contexts

A peer-reviewed Sensors paper defines Broadband over Power Lines as a PLC technology and discusses integrating Li-Fi, Wi-Fi, and BPL/PLC for indoor secure communications; it explicitly mentions RF, 220 V 50 Hz mains, and power-line infrastructure. ⁴³

A Huawei power-sector white paper discusses building a “digital twin of the entire power system” and separately notes that “wireless networks or power line communication” can connect low-voltage grids to users—explicitly bridging **digital twin + power line communication** in a utility context. ⁴⁴

The PLC/Li-Fi/digital twin sources reviewed did not also contain the SDA/DARPA/Blackjack constellation terms; they form a separate technology cluster (energy/industrial communications and digitalisation) rather than a directly evidenced space programme cluster. ⁴⁵

Jehovah’s Witnesses and Setecom are geographically anchored in Heredia but lack demonstrated linkage to the space and 5G programme chains

A Jehovah’s Witnesses official site page lists the Costa Rica contact address in Barreal de Heredia, Costa Rica, explicitly linking the organisation to the Heredia geography. ⁴⁶

Setecom's company contact page states it is located in Santo Domingo, Heredia, Costa Rica. ⁴⁷ Its LinkedIn description mentions generator controls, IP telemetry, and "DEEP SEA PLC," illustrating that "PLC" is used here in a vendor/branding sense and/or industrial-controls sense, not necessarily "power line communication." ⁴⁸

No reviewed sources explicitly connect Jehovah's Witnesses or Setecom to Kyndryl/IBM corporate operations, LeoLabs space tracking, SDA/DARPA programmes, or Costa Rica's 5G cybersecurity procurement rule beyond the shared geography of Heredia. ¹⁸

Source Catalogue and Cross-Reference Table

The table below lists sources that explicitly mention **combinations** of the user-specified terms. URLs are provided in code format to comply with output constraints.

Source	Date	Author / publisher	URL
Obtaining Frequency Response Measurements using an FFT	Not shown on PDF (Tech Note)	Audio Precision	<code>https://pearl-hifi.com/06_Lit_Archive/15_Mfrs_Publications/Audio_Precision/Freq_Resp_Measurements.pdf</code>
ffmpeg keep video/audio sync when using select filter	2016-08-24	Stack Overflow (answer by Gyan)	<code>https://stackoverflow.com/questions/39122287/ffmpeg-keep-video-audio-sync-when-using-select-filter</code>

Source	Date	Author / publisher	URL
Blackjack Focuses on Risk Reduction Flights and Simulations to Prepare for Full Demonstration	2020-05-11	DARPA	https://www.darpa.mil/news/2020/blackjack-risk-reduction
Blackjack Program Successfully Deploys Two Mandrake 2 Satellites	2021-07-07	DARPA	https://www.darpa.mil/news/2021/blackjack-deploys-mandrake
PWSA Tranche 0 fact sheet	2023 (document hosted by SDA)	SDA	https://www.sda.mil/wp-content/uploads/2023/04/Tranche-0-Fact-Sheet_FINAL.pdf
Transport Layer fact sheet	2024-05 (hosted by SDA)	SDA	https://www.sda.mil/wp-content/uploads/2024/05/Transport-Layer-Fact-Sheet_FINAL.pdf

Source	Date	Author / publisher	URL
Space Development Agency Successfully Launches Tranche 0 Satellites	2023-04-03	U.S. Space Force	https://www.spaceforce.mil/News/Article/3349198/space-development-agency-successfully-launches-tranche-0-satellites/
SMALL BUSINESS – SDA	Updated not shown	SDA	https://www.sda.mil/home/work-with-us/small-business/
Sedaro awarded Phase II SBIR supporting SDA and SSC	2024-09-12	Sedaro	https://www.sedaro.com/news/sedaro-awarded-phase-ii-sbir-supporting-sda-and-ssc
Sedaro portfolio entry (PWSA Digital Twin / DE Library)	Portfolio page (year tags shown)	SBIR.gov	https://www.sbir.gov/portfolio/1628131

Source	Date	Author / publisher	URL
LeoLabs announces Costa Rica Space Radar fully operational	2021-04-22	LeoLabs	https://www.leolabs.space/wp-content/uploads/2021/04/LeoLabs-Costa-Rica-Space-Radar_Final_4222021.pdf
LeoLabs Wins AFWERX Contract for Radar Technology	2024-06-13	Via Satellite	https://www.satellitetoday.com/government-military/2024/06/13/leolabs-wins-afwerx-contract-for-radar-technology/
Kyndryl country requirements invoicing page	Date not shown (retrieved 2026-03-10)	Kyndryl	https://www.kyndryl.com/us/en/procurement/supplier-resources/instructions/country-requirements
IBM and Kyndryl Client Guide to Operational Changes	2021-09-01 (PDF metadata shown in search)	IBM	https://www.ibm.com/services/us/kyndryl/kyndryl-billing-entitlement-information.pdf

Source	Date	Author / publisher	URL
Kyndryl launches innovation centre from Costa Rica	2021-11-04	CINDE	https://www.cinde.org/en/essential-news/kyndryl-launches-innovation-center-and-high-value-added-services-from-costa-rica
IBM expands security operations centre in Heredia	2019-04-04	Ministerio de Comercio Exterior (Costa Rica)	https://www.comex.go.cr/sala-de-prensa/comunicados/2019/abr-04-2019-ibm-consolida-costa-rica-como-experto-a-nivel-internacional/
Costa Rica 5G cybersecurity regulation index	2023-08-25 (decree date shown)	Costa Rica legal repository (PGR SCIJ)	https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo?nValor1=1&nValor2=100155&nValor3=137277&nValor5=2&param1=NRA
Costa Rica excludes China from 5G tech bidders	2023-09-08	Tico Times	https://ticotimes.net/2023/09/08/costa-rica-excludes-china-5g-tech-bidders

Source	Date	Author / publisher	URL
Costa Rica excludes 5G contracts to non-signatories	2023-09-06	Swissinfo (EFE wire)	https://www.swissinfo.ch/spa/costa-rica-excluye-de-contratos-pa%C3%ADses-no-signatarios-de-convenio-de-ciberseguridad/487
Costa Rican court suspends exclusion of Huawei	2024-02-14	RCR Wireless	https://www.rcrwireless.com/20240214/featured/costa-rican-court-suspends-exclusion-huawei-5g-provider
IBM GDPS qualification letter for Huawei DWDMs	2023-06-08 (page date shown)	IBM Support	https://www.ibm.com/support/pages/huawei-technology

Source	Date	Author / publisher	URL
Digitalization Trends in the Electric Power Industry 2030	2024 (PDF)	Huawei	https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/digitalization_trends_in_electric_power_industry_whitepaper_
Integrating Wi-Fi, Li-Fi, and BPL Technologies	2024	MDPI Sensors	https://www.mdpi.com/1424-8220/24/24/8105
Introducing Surya heliophysics foundation model	2025-08-20	IBM Research	https://research.ibm.com/blog/surya-heliophysics-ai-model-s

Source	Date	Author / publisher	URL
Expanded Prithvi model with global satellite data	2024-12-04 (updated 2025-02-26)	NASA Science	https://science.nasa.gov/science-research/ai-geospatial-mod
Costa Rica Jehovah's Witness contact page	Date not shown	Jehovah's Witnesses official site	https://www.jw.org/es/comunicados-internacionales/pa%C3%ADs-territorios/costa-rica/
SETECOM contact page	Date not shown	SETECOM S.A.	https://setecom.com/contacto
SETECOM LinkedIn page	Date not shown	LinkedIn company page	https://cr.linkedin.com/company/setecom

Source	Date	Author / publisher	URL
Kyndryl manufacturing page mentioning digital twins	Date not shown	Kyndryl	https://www.kyndryl.com/us/en/industries/manufacturing/transformative-technologies
Huawei opens service centres in Heredia	2019-10-27	Revista Estrategia y Negocios	https://www.revistaeyn.com/empresasymanagement/huawei-inaugura-nuevos-centros-de-servicio-en-costa-rica-NBEN1329895

Relationship and Timeline Visualizations

Relationship graph of the strongest explicit link chains

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graph TD
  A[46.875 Hz] --> B[48000/1024 DSP arithmetic]
  B --> C[FFT bin width at 48 kHz]
  B --> D[AAC 1024-sample frames @48 kHz]

  E[DARPA Blackjack] --> F[Mandrake 1]
  E --> G[Mandrake 2]
  G --> H[Optical inter-satellite links]
  E --> I[LEO constellation autonomy / mesh networking]

  J[SDA PWSA] --> K[Transport Layer LEO]
  K --> H
  K --> L[RF links]
  J --> M[Tranche 0 satellites]
  J --> N[Digital twin development interest]

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O[Sedaro] --> P[PWSA constellation digital twins]
 O --> Q[GPS satellite digital twins]
 P --> J
 Q --> R[GNSS test facilities]

S[LeoLabs] --> T[Costa Rica space radar]
 T --> U[LEO tracking / SSA]
 U --> V[Satellite operators / regulators]

W[Kyndryl] --> X[Heredia operations]
 Y[IBM] --> X
 X --> Z[America Free Zone / Free zones]

AA[Costa Rica 5G cybersecurity rule] --> AB[Procurement eligibility gate]
 AB --> AC[Huawei impact reported]
 AC --> AD[China referenced in news]

AE[PLC / BPL] --> AF[Li-Fi integration literature]
 AG[Huawei power whitepaper] --> AH[Power system digital twin]
 AG --> AE

AI[Jehovah's Witness address] --> AJ[Heredia]
 AK[Setecom] --> AJ

Timeline of high-signal dated events in the collected sources

timeline
 2019-04-04 : Costa Rica government communiqué: IBM expands security operations centre in Heredia
 2020-05-11 : DARPA describes Blackjack risk-reduction flights; names Mandrake 1 and Mandrake 2; references SDA partnership
 2021-04-22 : LeoLabs press release: Costa Rica Space Radar declared fully operational
 2021-11-04 : CINDE: Kyndryl completes separation from IBM and launches operations from Costa Rica
 2021-07-07 : DARPA: Mandrake 2 satellites deployed (Able/Baker); stakeholder team includes SDA
 2023-04-03 : U.S. Space Force: SDA successfully launches initial Tranche 0 satellites for PWSA
 2023-08-25 : Costa Rica legal repository: Executive Decree 44196 (5G cybersecurity regulation) dated
 2024-02-14 : Telecom trade press: court action reported suspending Huawei exclusion in 5G context
 2024-09-12 : Sedaro: SBIR awarded to build digital twins for GPS and PWSA constellation

Gaps, Caveats, and Next Steps

Key gaps and how to interpret them

The absence of credible sources tying **46.875 Hz** directly to satellites or LEO constellations is meaningful: across the primary technical note and independent technical discussion, it appears consistently as a **digital-audio/FFT framing artifact**. Any claim that 46.875 Hz is a space-system signature would need a *different* evidence trail, such as a satellite receiver design spec that explicitly chooses (for example) a 48 kHz sample rate and 1024-point processing window for RF demodulation and then ties that bin width to a specific operational waveform. None was found here. ¹⁷

The SDA/DARPA constellation chain is strong, but it is largely **U.S. defence architecture**; the Costa Rica 5G/Huawei chain is a **Costa Rica procurement and cybersecurity governance** topic; PLC/Li-Fi and digital twin sources often sit in **energy/industrial** contexts. In other words: multiple strong clusters exist, but explicit cross-cluster sources are limited. ⁵⁷

Follow-up queries designed to surface cross-cluster sources

To search for missing explicit bridges (especially **46.875 Hz ↔ satellites/LEO**), the most productive next step is to search for *architectures that use 48 kHz and 1024-sample windows for RF*, such as certain SDR pipelines, acoustic modem telemetry, or power-line comms equipment that may coexist with RF gear. Suggested queries (English and Spanish, and a Chinese option) are below; they are formatted to encourage retrieval of primary documentation:

"48000/1024" AND (satellite OR "software defined radio" OR SDR OR "ground station")

"46.875 Hz" AND (LEO OR "optical inter-satellite link" OR OISL OR "space mesh")

(site:.mil OR site:.gov) "48 kHz" "1024" AND (receiver OR demodulator OR FFT) AND (satellite OR "space")

"Decreto Ejecutivo 44196" "5G" "proveedores" "alto riesgo" "Huawei" (to find full decree PDF in official gazette mirrors)

"Zona Franca Los Arrallanes Edificio F-31" Kyndryl (to locate additional regulatory filings, permits, or procurement records)

"LeoLabs" "Costa Rica Space Radar" location coordinates (to retrieve land-use filings, environmental permits, or local press)

Spanish: "telemetría IP" "DEEP SEA PLC" Setecom "Santo Domingo Heredia" (to disambiguate PLC usage and identify technical suppliers)

Chinese: “数字孪生” “卫星星座” “SDA” “PWSA” (to find Chinese commentary that names SDA/PWSA explicitly, then back-link to program sources)

Practical next research steps

Locate full-text versions of Costa Rica’s 5G regulation as published in the official gazette (some official pages block automated access), and cross-check the decree’s operative clauses against media interpretations (vendor eligibility vs. procurement implementation). The official legal repository index is a stable anchor, but procurement effect is often established in implementing guidelines, tenders, or case law. ⁵⁸

For Kyndryl/IBM footprints in Heredia, pull company registry filings and free-zone programme lists (e.g., PROCOMER datasets) that may explicitly record name changes (IBM entities → Kyndryl) and operational category. In this pass, the most robust, directly quotable sources were Kyndryl’s own invoicing documentation and the IBM-hosted operational changes guide. ⁵⁹

For Huawei/IBM links, extend beyond qualification notices into standards participation, interoperability testing, and procurement qualification artefacts (e.g., telecom transport qualification letters like the IBM GDPS/Huawei DWDM notice). This is a stronger evidentiary path than secondary reporting about historical consulting relationships. ⁶⁰

¹ ³⁷ <https://www.comex.go.cr/sala-de-prensa/comunicados/2019/abril/cp-2353-ibm-consolida-costa-rica-como-experto-a-nivel-internacional/>

<https://www.comex.go.cr/sala-de-prensa/comunicados/2019/abril/cp-2353-ibm-consolida-costa-rica-como-experto-a-nivel-internacional/>

² ⁵ ¹³ ²⁷ https://www.sda.mil/wp-content/uploads/2023/04/Tranche-0-Fact-Sheet_FINAL.pdf

https://www.sda.mil/wp-content/uploads/2023/04/Tranche-0-Fact-Sheet_FINAL.pdf

³ ²⁵ <https://www.darpa.mil/news/2020/blackjack-risk-reduction>

<https://www.darpa.mil/news/2020/blackjack-risk-reduction>

⁴ ¹⁰ ¹⁴ ²¹ ³⁸ ⁵⁸ [https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?](https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA)

[nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA](https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA)

[https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?](https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA)

[nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA](https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_articulo.aspx?nValor1=1&nValor2=100155&nValor3=137277&nValor5=2¶m1=NRA)

⁶ ⁴⁴ [https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/](https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf)

[digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf](https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf)

[https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/](https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf)

[digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf](https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/digitalization_trends_in_electric_power_industry_whitepaper_2030_en.pdf)

⁷ ³⁰ <https://www.sda.mil/home/work-with-us/small-business/>

<https://www.sda.mil/home/work-with-us/small-business/>

⁸ ⁵⁶ <https://www.kyndryl.com/us/en/industries/manufacturing/transform-technologies>

<https://www.kyndryl.com/us/en/industries/manufacturing/transform-technologies>

⁹ ⁴³ ⁴⁵ <https://www.mdpi.com/1424-8220/24/24/8105>

<https://www.mdpi.com/1424-8220/24/24/8105>

- 11 18 46 <https://www.jw.org/es/comunicados-internacionales/pa%C3%ADses-territorios/costa-rica/>
<https://www.jw.org/es/comunicados-internacionales/pa%C3%ADses-territorios/costa-rica/>
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